

L1.3 HW/SW Platforms, Ethics Quiz ANS

1. How is automotive E/E architecture evolving according to the presentation?

- A) From about 80-100 distributed ECUs toward a few high-performance ECUs plus many small sensor/actuator-interface ECUs
- B) From centralized computing toward hundreds of independent high-performance computers
- C) From electronic control toward entirely mechanical control
- D) From wired ECUs toward eliminating all in-vehicle networks

ANS: A) From about 80-100 distributed ECUs toward a few high-performance ECUs plus many small sensor/actuator-interface ECUs

2. What is one benefit of consolidating automotive hardware into a few centralized ECUs?

- A) It eliminates the need for sensors.
- B) It simplifies the architecture, reduces network load, and can improve reliability.
- C) It guarantees zero power consumption.
- D) It increases the number of independent networks required.

ANS: B) It simplifies the architecture, reduces network load, and can improve reliability.

3. Which technology supports high-performance automotive computing with a small form factor and low power consumption?

- A) Mechanical relays
- B) A trunk full of desktop computers
- C) System-on-Chip (SoC) technology
- D) An external data center as the only processor

ANS: C) System-on-Chip (SoC) technology

4. Why is hard real-time operation a design objective for autonomous vehicles?

- A) It reduces the number of vehicle doors.
- B) It guarantees that every algorithm uses a GPU.
- C) It makes software updates unnecessary.
- D) A missed deadline may compromise safety.

ANS: D) A missed deadline may compromise safety.

5. What is identified as the most compute-intensive AV workload?

- A) Deep-learning inference

- B) Door-lock control
- C) Audio playback
- D) Windshield-wiper control

ANS: A) Deep-learning inference

6. What is a typical automotive use of a Digital Signal Processor (DSP)?

- A) Global route planning
- B) Image preprocessing
- C) Mechanical braking
- D) Storing an HD map on paper

ANS: B) Image preprocessing

7. Why are GPUs well suited to deep-learning workloads?

- A) They execute only one operation at a time.
- B) They contain no arithmetic units.
- C) They have many computational elements for highly parallel operations such as matrix multiplication.
- D) They are designed exclusively for body electronics.

ANS: C) They have many computational elements for highly parallel operations such as matrix multiplication.

8. What is NVDLA in the NVIDIA DRIVE platform?

- A) An automotive Ethernet protocol
- B) A mechanical lidar scanner
- C) A general-purpose operating system
- D) An ASIC for deep-learning inference

ANS: D) An ASIC for deep-learning inference

9. What is an FPGA?

- A) Reprogrammable hardware made of configurable logic blocks and programmable interconnections
- B) A fixed-function mechanical controller
- C) A satellite-based localization service
- D) A multimedia network protocol

ANS: A) Reprogrammable hardware made of configurable logic blocks and programmable interconnections

10. What is a drawback of using a large GPU batch for inference?

- A) It always lowers throughput.
- B) It can produce high and nondeterministic latency for an individual data item.
- C) It prevents SIMD computation.
- D) It converts the GPU into an FPGA.

ANS: B) It can produce high and nondeterministic latency for an individual data item.

11. What is a key advantage of batch-less FPGA inference?

- A) It requires very large batches.
- B) It removes all hardware constraints.
- C) It can provide low and deterministic latency for any batch size.
- D) It is used only for neural-network training.

ANS: C) It can provide low and deterministic latency for any batch size.

12. Where can automotive FPGAs be deployed?

- A) Only in cloud data centers
- B) Only in infotainment displays
- C) Only in mechanical engine parts
- D) In smart sensors or as a central compute engine

ANS: D) In smart sensors or as a central compute engine

13. What are ASICs for deep learning often called?

- A) Neural Processing Units or AI accelerators
- B) Local Interconnect Networks
- C) Flash bootloaders
- D) Time-Sensitive Networks

ANS: A) Neural Processing Units or AI accelerators

14. What is a major drawback of an ASIC compared with an FPGA?

- A) Lower specialization for its target computation
- B) Loss of programmability and flexibility
- C) A requirement to execute from a hard disk
- D) Inability to perform deep-learning inference

ANS: B) Loss of programmability and flexibility

15. Which platform was used for Tesla Autopilot HW1 from approximately 2014 to 2016?

- A) NVIDIA DRIVE PX2

- B) Tesla FSD HW4
- C) Mobileye EyeQ3
- D) Bosch AI Car Computer

ANS: C) Mobileye EyeQ3

16. Which platform was used for Tesla Autopilot HW2?

- A) Mobileye EyeQ3
- B) Tesla HW4 FSD processor
- C) Google TPU
- D) NVIDIA DRIVE PX2

ANS: D) NVIDIA DRIVE PX2

17. Which sensor configuration is listed for Tesla HW4?

- A) Eight cameras
- B) Eight cameras, one radar, and twelve ultrasonic sensors
- C) Two cameras, one radar, and twelve ultrasonic sensors
- D) Six lidars and four radars

ANS: A) Eight cameras

18. How many CPU cores are listed for the Tesla HW3 FSD processor?

- A) 8
- B) 12
- C) 20
- D) 36

ANS: B) 12

19. Which NPU configuration is listed for Tesla HW4?

- A) One NPU at 10 TOPS
- B) Two NPUs totaling 36 TOPS
- C) Three NPUs totaling 50 TOPS
- D) No NPUs

ANS: C) Three NPUs totaling 50 TOPS

20. Which processors form the basis of the Delphi/Audi zFAS integration solution?

- A) Only a Google TPU
- B) Tesla HW3 and HW4 processors
- C) AMD Ryzen and Intel Xeon processors

D) NVIDIA Tegra K1 and Mobileye EyeQ3

ANS: D) NVIDIA Tegra K1 and Mobileye EyeQ3

21. Which in-vehicle network is positioned as the high-bandwidth backbone?

- A) Ethernet
- B) LIN
- C) Low-speed CAN
- D) MOST only

ANS: A) Ethernet

22. Which network is intended primarily for multimedia audio/video transmission?

- A) FlexRay
- B) MOST
- C) LIN
- D) Low-speed CAN

ANS: B) MOST

23. Which network is associated with safety-critical X-by-Wire applications?

- A) MOST
- B) LIN
- C) FlexRay
- D) Bluetooth

ANS: C) FlexRay

24. Which network is used for body electronics such as doors, lights, and rearview mirrors?

- A) Ethernet TSN
- B) MOST150
- C) FlexRay
- D) LIN

ANS: D) LIN

25. What data rate is listed for CAN-FD?

- A) 8 Mbps
- B) 10 kbps

- C) 150 Mbps
- D) 10 Gbps

ANS: A) 8 Mbps

26. When was the AUTOSAR development partnership founded?

- A) 1995
- B) 2003
- C) 2015
- D) 2023

ANS: B) 2003

27. For what type of ECU is AUTOSAR Classic Platform intended?

- A) Cloud-only servers
- B) Non-safety-critical smartphones
- C) Resource-constrained, safety-critical ECUs
- D) High-performance gaming computers

ANS: C) Resource-constrained, safety-critical ECUs

28. For what type of ECU is AUTOSAR Adaptive Platform intended?

- A) Mechanical control systems with no software
- B) Only low-speed body-electronics nodes
- C) Resource-constrained 8-bit controllers only
- D) High-performance multicore automotive ECUs

ANS: D) High-performance multicore automotive ECUs

29. Which operating system standard is associated with AUTOSAR Classic?

- A) OSEK
- B) POSIX
- C) Android
- D) DDS

ANS: A) OSEK

30. Which communication style is associated with AUTOSAR Adaptive?

- A) Statically configured signal-based communication only
- B) Dynamically configured service-oriented communication using SOME/IP
- C) Mechanical point-to-point signaling
- D) ROS 1 Master-only communication

ANS: B) Dynamically configured service-oriented communication using SOME/IP

31. What type of real-time constraint is listed for AUTOSAR Classic?

- A) No real-time constraint
- B) Best-effort only
- C) Hard real-time
- D) Soft real-time only

ANS: C) Hard real-time

32. Which platform should remain close to deterministic actuator and low-level vehicle-control functions?

- A) Android
- B) ROS 2 research tools
- C) AUTOSAR Adaptive
- D) AUTOSAR Classic

ANS: D) AUTOSAR Classic

33. What service does the ROS 1 Master node provide?

- A) Naming and registration services for other nodes
- B) GPU scheduling for CUDA kernels
- C) Flash-memory rewriting
- D) Vehicle-to-cloud routing

ANS: A) Naming and registration services for other nodes

34. What middleware does ROS 2 use for publish-subscribe communication?

- A) CAN-FD
- B) DDS
- C) MOST150
- D) OSEK

ANS: B) DDS

35. Why is standard ROS difficult to certify to a high automotive safety level?

- A) It cannot communicate between nodes.
- B) It contains no software libraries.
- C) It relies on Linux as its underlying operating system.
- D) It requires AUTOSAR Classic for every message.

ANS: C) It relies on Linux as its underlying operating system.

36. To what Automotive Safety Integrity Level is the Apex.ai framework described as certified as a Safety Element out of Context?

- A) ASIL-A
- B) ASIL-B
- C) ASIL-C
- D) ASIL-D

ANS: D) ASIL-D

37. Which NVIDIA DRIVE component is the foundational stack containing an embedded RTOS, hypervisor, CUDA libraries, and TensorRT?

- A) DRIVE OS
- B) DRIVE IX
- C) DRIVE AV
- D) DriveWorks application only

ANS: A) DRIVE OS

38. Which NVIDIA DRIVE component provides perception, mapping, and planning modules?

- A) DRIVE IX
- B) DRIVE AV
- C) MOST150
- D) AUTOSAR RTE

ANS: B) DRIVE AV

39. On which ROS version is Autoware.AI based?

- A) DDS without ROS
- B) ROS 2
- C) ROS 1
- D) AUTOSAR Adaptive

ANS: C) ROS 1

40. Which statement describes Baidu Apollo?

- A) A proprietary platform available only on NVIDIA hardware
- B) A Japanese platform based exclusively on ROS 1
- C) A body-electronics network protocol
- D) An open-source, hardware-neutral autonomous-driving platform from China

ANS: D) An open-source, hardware-neutral autonomous-driving platform from China

41. What is the purpose of a flash bootloader in an ECU?

- A) To erase and rewrite flash memory for programming or software updates
- B) To accelerate matrix multiplication
- C) To transmit multimedia over MOST
- D) To perform object detection from cameras

ANS: A) To erase and rewrite flash memory for programming or software updates

42. In an OTA update using a Connectivity ECU, when is the update loaded to the target ECU?

- A) Continuously while the vehicle is driving at full speed
- B) When the vehicle is in a safe state
- C) Only when all ECUs are powered off
- D) Before the software is wirelessly transferred

ANS: B) When the vehicle is in a safe state

43. Where is new software stored during an OTA update without a Connectivity ECU?

- A) Only in the driving application's active memory
- B) On a remote server with no local copy
- C) In a memory area separate from the driving application
- D) Inside an ultrasonic sensor

ANS: C) In a memory area separate from the driving application

44. In the classic trolley problem, what happens if the lever is pulled?

- A) The trolley stops with no casualties.
- B) The five people remain in danger and the side track stays unused.
- C) All six people are saved.
- D) The trolley moves to the side track, where one person is killed instead of five on the main track.

ANS: D) The trolley moves to the side track, where one person is killed instead of five on the main track.

45. Approximately how many decisions did the MIT Moral Machine platform gather?

- A) 40 million
- B) 400 thousand
- C) 4 billion
- D) 40 thousand

ANS: A) 40 million

46. How many factors were used to generate Moral Machine accident scenarios?

- A) Three
- B) Nine
- C) Twelve
- D) Twenty

ANS: B) Nine

47. How many large cultural clusters are identified in the Moral Machine results presented?

- A) Two
- B) Four
- C) Three
- D) Nine

ANS: C) Three

48. Which preference was weaker in the Southern cultural cluster than in the other two clusters?

- A) Sparing lawful pedestrians over jaywalkers
- B) Sparing younger people over older people
- C) Sparing pedestrians over passengers
- D) Sparing humans over pets

ANS: D) Sparing humans over pets

49. According to the presentation, what caused recent AV accidents rather than trolley-problem-style ethical choices?

- A) Failures in sensors or perception algorithms
- B) Deliberate choices to sacrifice passengers
- C) Cultural preferences programmed into the vehicle
- D) Moral Machine survey responses

ANS: A) Failures in sensors or perception algorithms

50. What does the presentation state about U.S. legislation for public-road AV testing?

- A) Every state has identical requirements.
- B) Regulation differs widely across states.
- C) Public-road testing is prohibited nationwide.
- D) Only federal law applies, with no state role.

ANS: B) Regulation differs widely across states.